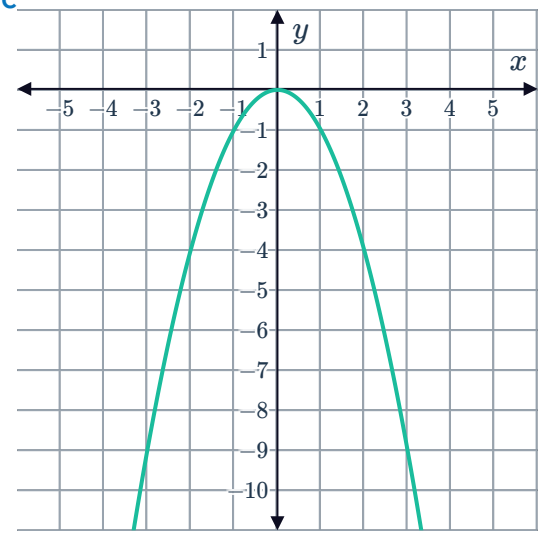
Finding the equation of a parabola

1

a $a = -1$

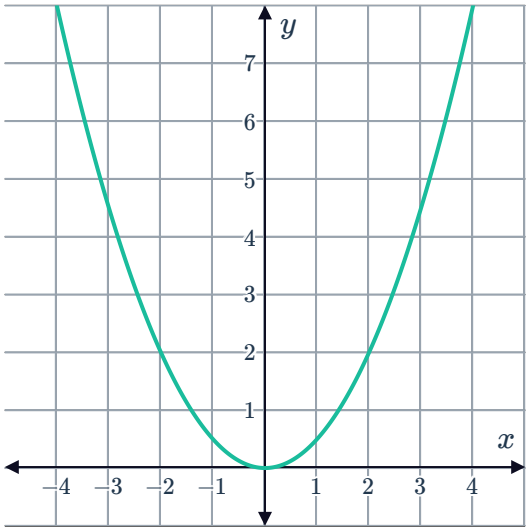
b $(0, 0)$

c



2

a



b $y = \frac{1}{2}x^2$

3

a No

b Yes

4

a $(3, 5)$

b $y = (x - 3)^2 + 5$

5

a $x = -7$

b $y = -\frac{3}{7}(x - 1)(x + 7)$

c $\left(-3, \frac{48}{7}\right)$

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7

a $x = -2$

b $y = (x+2)^2 - 9$

8 $y = x^2 + 6x - 40$

9

a $x^2 - b^2$

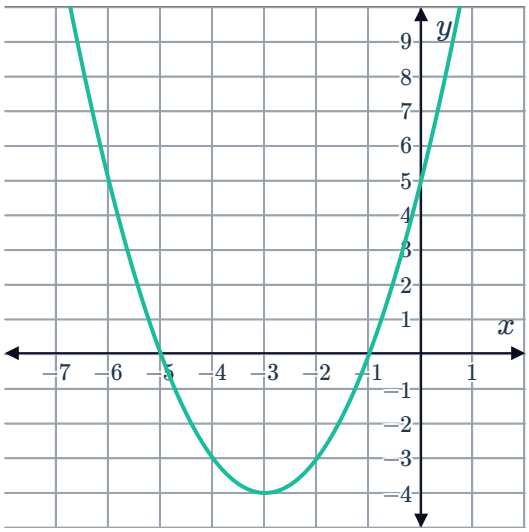
b $a = 0$

10

a $y = (x+1)(x+5)$

b $y = 5$

c

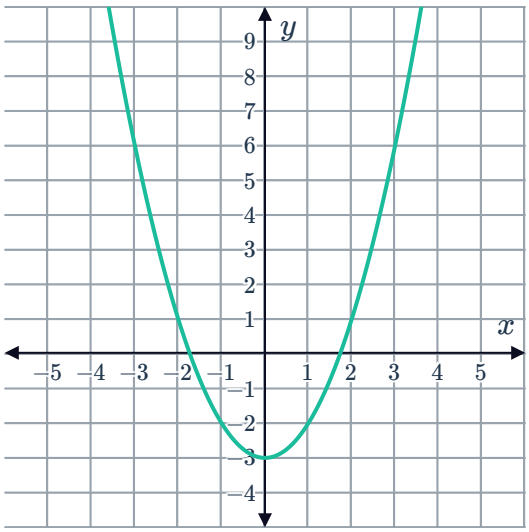


11

a $y = x^2 - 3$

b -2

c



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12

a infinitely many

b infinitely many

c 1

d $y = -\frac{3x^2}{25} - \frac{6x}{25} + \frac{72}{25}$

13 $y = \frac{x^2}{4} + 6x + 39$

14 $b = 4$

15 $c = 4$

16 $k = \frac{1}{3}$

17

a $a + b + c = 3, 9a + 3b + c = -1, 16a + 4b + c = 0$

b $c = 8$

c $a = 1$

d $b = -6$

e $x^2 - 6x + 8$

Applications

18

a Negative a, positive c

b $c = 2560$

c $a = -\frac{1}{10}$

d $\frac{25\,071}{10}$ m

19

a $y = -\frac{1}{20}(x - 18)^2 + 16.2$

b $4071.50\ \text{m}^2$

20

a $A(0, 2.5), B(2, 0), C(6, 0)$

b 2

c $4a + k = \frac{5}{2}$

d $0 = 16a + k$

e $a = -\frac{5}{24}$

f $k = \frac{10}{3}$

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21

a $A(0, 3.5), B(3, 0), C(7, 0)$

b 3

c $9a + k = \frac{7}{2}$

d $0 = 16a + k$

e $a = -\frac{1}{2}$

f $k = 8$

22

a $(180, 98)$

b $y = -\frac{49}{16\,200}(x - 180)^2 + 98$

c Below

23

a $A(18, 0), B(0, 108)$

b $y = -\frac{1}{3}(x - 18)(x + 18)$

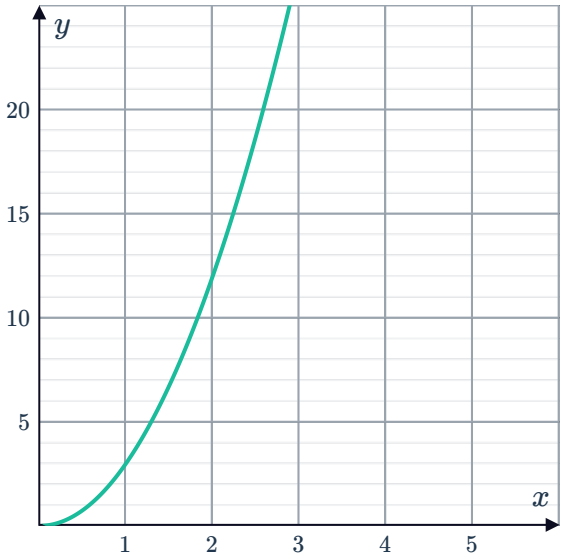
c $x = 9\sqrt{2}$

24

a

x	0	1	2	3	4
y	0	3	12	27	48

b



c 0 metres

d 0

e $(0, 0)$

25

a Initially, the object has not fallen any distance and is 700 m above the ground.

b decreasing

c

i True **ii** True **iii** False

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26

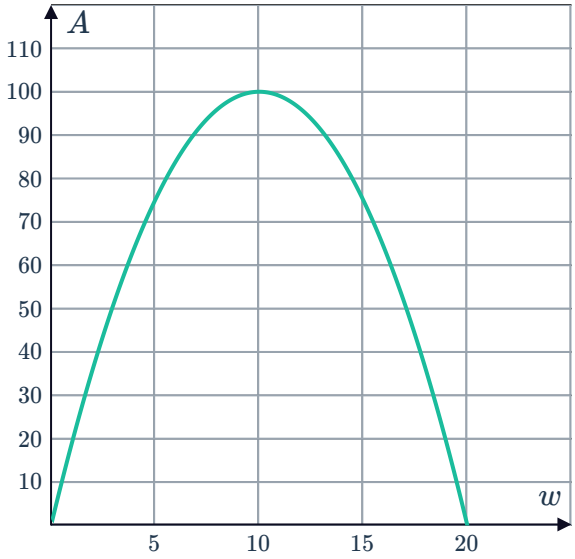
a After 4.5 seconds, the pebble is 40.5 metres above the ground. The highest distance the pebble reaches above the ground is 40.5 metres.

b $t = 0, 9$

c The times when the pebble is on the ground.

27

a



b $w = 0, 20$

c There is no corresponding rectangle because its area would be 0.

d 100

28

a $0, 40$

b 20

c The largest rectangular area that can be fenced off with 80 m of wire.

29

a $s = 12$ **b** 337

30

a $t = 20$ **b** 100 metres

31

a $13 - x$ **b** $x\,(13 - x)$ **c** 6.5 **d** 42.25 m^2

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Applications using graphing calculators

32

a $(0, 0)$

b The initial distance fallen by the object.

33 $-14,14 \text{ m/s}$

34

a 176.4 m

b 6 s

c 12 s

35

a 900 m

b Initially, the object has not fallen any distance.

c 13.55 s

d After 13.55 seconds the object is on the ground.

36

a $0, 30$

b 15

c The largest rectangular area that can be fenced off with 60 m of wire.

37

a 4 s

b 10 s

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